



B9AI103 Recommender Systems

CA TWO

Assessment:	3
Assignment Task	4
Overall marks will be awarded according to the following criteria:	4
Resources	5
Grading Criteria	6

Assessment:

Item	Description
Module Title:	Recommender Systems
Module Code:	B9AI103
Module Lecturer (s):	Nitya Govindaraju
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	CA TWO
Percentage (%) Weighting:	40%
MIMLOs being assessed:	3,4,6
Assessment Number:	2
Individual/Group:	Individual
Issue Date:	Week 7
Submission Date:	Week 13
Feedback Date:	Week 15
Feedback Strategy:	Moodle

Assignment Task

Individual Report on Explainability and Ethics in Recommender Systems

Each student will produce an in-depth individual report that combines practical application of Explainable AI (XAI) with critical assessment of the ethical and social implications of a recommender system

Overall marks will be awarded according to the following criteria:

Task	Description	Marks	MIMLO Alignment
Part 1: XAI Implementation and Explanation	Select an existing recommender system model or a relevant dataset. Apply two distinct XAI methods (e.g., SHAP, LIME, or Google's What-If Tool) to generate and compare explanations for specific recommendations. Detail the XAI methodology chosen and its effectiveness in creating transparent, user-centric explanations.	50	4
Part 2: Ethical Analysis and Mitigation Strategy	Based on the system analyzed (in Part 1 or another case study), critically assess a key ethical or social implication, such as bias (echo chambers, popularity bias), fairness, or privacy. Propose and detail at least two mitigation strategies (e.g., bias detection, federated learning) for the identified issue.	35	3
Part 3: Generative AI in the Ethical Context	Critically evaluate the ethical implications of using Generative AI in the context of the recommender system you have studied. Discuss how the introduction of generative models might complicate or assist in addressing the ethical challenge identified in Part 2, including any challenges for explainability	15	6
Total		100	3,4,6

Deliverables:

- Individual Analytical Report (Max 2,500 words): A formal academic report that details the chosen system, XAI application, results, critical ethical assessment, mitigation strategies, and the Generative AI discussion.
- Code/Notebook: Include a properly documented code (e.g., Jupyter Notebook) demonstrating the application of the XAI tools (SHAP/LIME).

Resources

Software or tools required/ useful:

- Programming Languages: Python
- Libraries/Frameworks: TensorFlow, PyTorch, Scikit-learn, Surprise, SHAP, LIME, Pandas, NumPy
- Data Visualization: Matplotlib, Seaborn
- Collaboration Tools: GitHub, Jupyter Notebook

Templates, files, data tables provided by the lecturer:

- Project template for documenting group projects
- Sample datasets for training and evaluation
- Evaluation metrics sheet

Highlight the Appropriate Selection as per the programme Module and Assessment Document

1	2	3	4	5
NO AI	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI-ASSISTED EDITING	AI TASK COMPLETION, HUMAN EVALUATION	FULL AI
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated

Grading Criteria		Evaluative Criteria/Performance Level Descriptors (what does performance at this level look like?)			
Scale	Grade Band	Criterion 1 – XAI Application & Analysis (50 Marks)	Criterion 2 – Ethical Assessment & Mitigation (38 Marks)	Criterion 3 – GenAI Ethical Critique (12 Marks)	Criterion 4 – Report Quality & Rigor
Exceptional	90-100%	Masterful application of two complex XAI techniques. Analysis is profound , critical, and theoretically well-supported.	Analysis of ethical issues is profound , supported by academic literature. Mitigation strategies are innovative, practical, and highly defensible.	Provides an original and sophisticated critical evaluation of the ethical risks and benefits introduced by using GenAI.	Flawlessly structured, highly professional, perfectly referenced, and exceeds academic standards.
Outstanding	80-89%	Flawless application and excellent comparison of the two required XAI methods. Demonstrates deep understanding of interpretability goals.	Highly detailed and critical assessment of a major ethical issue, linked explicitly to system mechanics. Mitigation strategies are strong, well-justified, and realistic.	Thoroughly critiques the ethical and explainability challenges presented by GenAI in the recommender space.	Excellent structure and clarity. Referencing is accurate and presentation is highly professional.
Excellent	70-79%	Competent and correct application of two XAI methods. Clearly explains the XAI results and their implications for transparency.	A clear, detailed assessment of an ethical issue. Proposes two appropriate and well-justified mitigation strategies.	Clearly articulates how GenAI complicates or assists in addressing the ethical challenge.	Report is well-structured and written. Referencing is consistently correct.

Very Good	60-69%	Good application of the XAI methods, but the analysis of the results may be more descriptive than critical .	Identifies and assesses an ethical issue, though the discussion may rely heavily on general principles. Proposed mitigation strategies are appropriate but lack depth.	Provides a good general discussion of GenAI's ethical role in recommendations.	Report is logical and professional. Minor errors in structure or referencing may be present.
Good	50-59%	Correctly attempts to apply two XAI methods, though minor technical or interpretive errors may be present. Good foundational understanding .	Identifies a relevant ethical issue, but the assessment is brief or superficial. Proposed mitigation strategies are generic or not fully linked to the specific problem.	Includes a basic discussion of Generative AI's ethical dimension.	Report meets the structural and academic requirements but lacks depth in argumentation.
Sufficient	40-49%	Minimal attempt to apply XAI, or only one method is demonstrated, with significant errors in execution or interpretation.	Identifies an ethical concern but fails to critically assess its implications or propose a mitigation strategy.	GenAI critique is weak or superficial.	Report is poorly structured, contains multiple errors, or referencing is inconsistent.
Somewhat/ Unfulfilled	30-39%	Severe errors in XAI application; output is difficult to interpret or demonstrates a lack of core knowledge .	Assessment is vague, irrelevant, or fails to address both ethical assessment and mitigation.	The Generative AI discussion is extremely brief or irrelevant.	Report is poorly presented and difficult to follow.
Unfulfilled	0-29%	XAI application is entirely	Fails to identify or assess an ethical	The Generative AI discussion is	The assessment is incomplete

		missing or non-functional. Demonstrates a lack of understanding of required techniques.	implication, or the discussion is entirely absent.	absent.	or fails to meet the minimum requirements.
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)					
MIMLO to which criterion is aligned:		MIMLO 4	MIMLO 3	MIMLO 6	MIMLO 3, 4, 6

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
 - c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).
 - a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's Generative AI Guidelines HERE for further information.
 - c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
 - a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.
 - d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.

- e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)
6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.